

# BEHRAD SADEGHI

AI/ML Engineer | MSc Data Science

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## SKILLS

- Programming:** Python, SQL, R
- Machine Learning:** PyTorch, Scikit-learn, Transformers, Natural Language Processing (NLP), Graph Neural Networks (GNNs)
- LLM & GenAI:** Vertex AI, Gemini, RAG, LangGraph, Google Agent Development Kit (ADK), Crew AI, Prompt Engineering
- Cloud & Infrastructure:** Google Cloud Platform (GCP), Cloud Run, BigQuery, Firestore, Docker, Terraform, FastAPI, Git
- Data:** Pandas, NumPy, Matplotlib, Seaborn, MongoDB
- Languages:** English (IELTS 7.0), Italian (Elementary), Persian (Native)

## WORK EXPERIENCE

**AI Engineer** May 2026 – Present  
*Softlab S.p.A.* Milan, Italy

- Architected an automated monitoring platform for a **7-stage AI pipeline**, computing KPI metrics from **Firestore** and surfacing root-cause failures from noisy **Cloud Logging** output — deployed as a scheduled, **Terraform**-managed **Cloud Run Job**.
- Designed and implemented an end-to-end **Vertex AI** batch inference pipeline that automatically summarized and classified previously unprocessed customer calls using **Gemini** structured outputs, eliminating manual first-pass triage while achieving **94% agreement** with client-validated classifications and **zero parsing failures**.
- Engineered strict, campaign-specific response schemas that constrain **Gemini** outputs to a fixed set of valid categories, eliminating hallucinated or malformed model outputs across concurrent **Vertex AI** batch jobs — all within a fully configuration-driven architecture that lets new campaigns launch without modifying prompts or model code.

**Machine Learning Engineer — Fraud Detection** Sep 2024 – Oct 2025  
*SnappBox* Tehran, Iran

- Designed and deployed an end-to-end fraud detection system processing **thousands of ride events daily**, serving real-time inference via **FastAPI** REST APIs on **Docker** and **Google Cloud**. Engineered scalable feature pipelines in **BigQuery**, reducing false positives by **20%** compared to the previous rule-based system.
- Developed machine learning models to detect anomalous rider behavior and coordinated fraud patterns, improving detection precision by **15%**. Designed behavioral fraud indicators that helped avert an estimated **15–20%** of fraud losses.

**Machine Learning Developer Intern** Feb 2023 – Aug 2023  
*Infinite Modern Technology* Tehran, Iran

- Processed and analyzed real-world datasets using **Pandas**, **NumPy**, and **Seaborn**; built automated **EDA pipelines** that reduced manual reporting time and surfaced actionable insights for business stakeholders.
- Trained and evaluated **supervised and unsupervised ML models** (classification, clustering) using Scikit-learn; tuned hyperparameters and validated results through cross-validation to support data-driven product decisions.

**Research Assistant** Feb 2022 – Jan 2023  
*University of Guilan* Rasht, Iran

- Led development of a **cross-attention-based model** for **Nasal Fracture Detection** on a novel, previously unexplored clinical dataset.

## SELECTED PROJECTS

**Transaction Fraud Detection Using GNNs and Tabular Models** | [GitHub](#) PyTorch / GNNs / Transformers

- Built a fraud detection system combining supervised (**Tabular Transformers**, **NNs**) and semi-supervised (**GraphSAGE**, **GAT**) models. Engineered a **hybrid loss function** for graph-based learning with only **15% labeled data**, achieving competitive F1 scores under realistic label scarcity.

**FraudRide-Analytics** | [GitHub](#) Pandas / Scipy / Seaborn

- Analytics pipeline for detecting **fraudulent bikers** in ride-sharing. Applied **statistical hypothesis testing** and **anomaly detection** to identify suspicious trip patterns, simulating production-level fraud analysis for mobility platforms.

**Essay Agent** | [GitHub](#) LangGraph / LangChain / Streamlit

- Agentic essay-writing assistant built with **LangGraph** and **Streamlit**, guiding users through the full cycle of **planning**, **researching**, **drafting**, and **critiquing** using multi-step LLM orchestration.

## EDUCATION

**University of Milan** Sep 2025 – Present  
*MSc in Data Science for Economics and Health* Milan, Italy

**University of Guilan** Sep 2019 – Feb 2024  
*BSc in Computer Engineering* Rasht, Iran